# Numerical Analysis – Homework #3

Utah State University Fall 2026

**1**.

Consider the system of linear equations:

$$\begin{aligned} x_1 + 2x_2 &= -1, \\ -2x_1 - 3x_2 + 2x_3 &= 3, \\ -2x_2 - 3x_3 &= -3. \end{aligned}$$

(a) Write the system as $Ax = b$.

(b) Solve the system using Gaussian Elimination.

(c) Find the $LU$ decomposition of $A$ and solve using it.

(d) Compute $A^{-1}$.

(e) What is the determinant of $A$?

(f) What is the condition number of $A$?

**2**.

Evaluate the norms $\|*\|_p$ where $p = 1, \infty$ for the following.

(a)

$$A = \begin{bmatrix} -9 & 5 & -9 \\ -2 & 7 & 5 \\ 5 & 1 & 8 \end{bmatrix}$$

(b)

$$B = \begin{bmatrix} 8 & -2 & 1 \\ -2 & 2 & -1 \\ -2 & 4 & -3 \end{bmatrix}$$

(c) Compute $B^2$, $A + B$, and $AB$. Does the triangle inequality hold?

**3**.

Let $A$ and $B$ be two $n \times n$ matrices. Show that the number of nonzero scalar multiplications required to compute $C = AB$ is (using Matlab notation)

$$\sum_{i=1}^{n} \mathrm{nnz}\big(A(:,i)\big) * \mathrm{nnz}\big(B(i,:)\big).$$

**4**.

Consider

$$A(\theta) = \begin{pmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{pmatrix}.$$

(a) Write $\kappa_1\big(A(\theta)\big) \equiv \|A\|_1 \, \|A^{-1}\|_1$ as a function of $\theta$ for $\theta$ near zero.

(b) Suppose $\theta = 10^{-5}$ and we solve $A(\theta)x = b$ using Gaussian elimination with partial pivoting on a machine that carries sixteen decimal digits of precision. According to the rule of thumb given in class, what is a relative bound on the error in the computed solution and about how many decimal digits of accuracy can we expect?

**5**.

Suppose you want to find $x$ satisfying $Ax = b$, where

$$A = \begin{pmatrix} 1.01 & 1 \\ 1 & 1.01 \end{pmatrix}.$$

Instead of having $b$ exactly, you have only a vector $\hat{b}$ obtained through experimental observations, the components of which have only about three digits of accuracy. If $\hat{x}$ satisfies $A\hat{x} = \hat{b}$, then find an upper bound on $\|\hat{x} - x\|/\|x\|$ using our perturbation theorem. You may use any norm you like and any means you like (MATLAB or otherwise) to evaluate norms and $\kappa(A)$, but explain what you do.

*Note:* If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \text{then} \quad A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

**6**.

Suppose $A$ is an $n \times n$ matrix and $b$ is an $n$-vector given by

$$A = \begin{pmatrix} 1 & 0 & \cdots & 0 & 1 \\ -1 & 1 & \ddots & \vdots & \vdots \\ \vdots & -1 & \ddots & 0 & 1 \\ \vdots & \vdots & \ddots & 1 & 1 \\ -1 & -1 & \cdots & -1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ \vdots \\ \vdots \\ 1 \\ 2 \\ 1 \end{pmatrix}.$$

Applying Gaussian elimination with partial pivoting to $Ax = b$ gives $Ux = c$, where

$$U = \begin{pmatrix} 1 & 0 & \cdots & 0 & 1 \\ 0 & 1 & \ddots & \vdots & 2 \\ \vdots & 0 & \ddots & 0 & \vdots \\ \vdots & \vdots & \ddots & 1 & 2^{n-2} \\ 0 & 0 & \cdots & 0 & 2^{n-1} \end{pmatrix}, \qquad c = \begin{pmatrix} 1 \\ 2 \\ \vdots \\ 2^{n-3} \\ 2^{n-2} + 1 \\ 2^{n-1} + 1 \end{pmatrix}.$$

(Incidentally, it turns out there is *no* pivoting, but this isn't important for the problem.) What solution will be computed in finite precision arithmetic when $2^{n-2} > 1/\epsilon$, where $\epsilon$ is machine epsilon (unit roundoff error)? Feel free to experiment using MATLAB, but explain your answer. *(Remember: $\epsilon$ can be computed using MATLAB's* `eps` *command.)*